TrailBuddy: experimental protocol

Expanded tool list | 3 experiment options | measurement reasoning

Core question: Do trekking poles reduce estimated knee extensor demand during descent, and does their length change the demands placed on the arms?

Study scope: 3 team members as participants. Within-person pilot only, not proof of injury prevention or a population-wide optimal pole length.

1 GitHub tools we can actually use

Start with the first two for a fast video pilot. Add force measurements and modeling only after the camera setup works.

Repository	What it contributes	Important limit
mediapipe2dangle[GitHub]	Easy video → hip/knee angles + CSV	Simplest starting code; 2D angles only.
Sports2D[GitHub]	Phone video → body angles and motion files	Better full-body tracking; assumes mostly planar motion.
OpenCap Core[GitHub]	2+ phones → 3D pose and joint angles	More setup; poles can hide body landmarks.
OpenCap Processing[GitHub]	Process OpenCap motion; run muscle-driven models	Model-derived forces, not direct measurements; validate pole contact assumptions.
OpenSim Core[GitHub]	Musculoskeletal model and inverse dynamics	Best for advanced force estimates; requires suitable external-force inputs.
Pose2Sim[GitHub]	2+ camera 3D tracking → OpenSim motion	Flexible phones/cameras; needs calibration and synchronization.
pyomeca[GitHub]	Clean, filter, synchronize, and visualize data	Useful glue for CSV/TRC/MOT files; does not supply forces.
FreeMoCap[GitHub]	Multicamera 3D markerless capture	Alternative if we can set up USB webcams.
Kinovea[GitHub]	Manually check angles and gait events	Good quality-control check for AI pose tracking.
HX711 Arduino library[GitHub]	Read a calibrated load-cell signal	Possible DIY pole-force sensor; hardware and safety validation needed.
AnyBody AMMR[GitHub]	Detailed knee/arm musculoskeletal model library	Requires licensed AnyBody system; not a free modeling engine.

Most promising combination: Sports2D + pyomeca for rapid kinematics; OpenCap + OpenCap Processing for modeled dynamics; OpenSim/AnyBody only after external-force assumptions are clear.

2 How we could combine tools using AI

We do not need to write a new motion-capture system. AI can help connect the existing repositories with short Python scripts, convert exports, and make repeatable plots.

PATH A | Fastest Phone video → Sports2D / mediapipe2dangle → CSV → pyomeca / Python → angles + gait-event plots

PATH B | Better 3D Two phone videos → OpenCap → OpenCap Processing → joint motion + tentative modeled muscle demand

PATH C | Force-focused Video + foot force (+ calibrated pole force) → OpenSim or a stated 2D model → knee extensor-force estimate

AI can: write import/synchronization scripts, flag missing tracking points, calculate angles, and make figures. AI cannot: create missing pole forces, guarantee accurate joint loads, or validate a biomechanical model.

3 Three experiment options for our three-person team

OPTION 1 | Simple stair comparison [best for Nov 17]

3 people × 2 conditions × 3 descents = 18 trials

Test: Use CREATE lab steps. Compare no poles vs regular-length poles, adjusted to ~90° elbow flexion on level ground. Film from the side; keep pace and starting foot similar.

Output: Knee and shoulder angles, trunk lean, descent time; model knee force only if a defensible force estimate is available.

Trade-off: Fast and repeatable, but stairs are not a downhill trail.

OPTION 2 | Does pole length matter?

3 people × 4 conditions × 3 descents = 36 trials

Test: Same steps and planting technique. Compare no poles, normal length, normal −5 cm, and normal +5 cm. Randomize condition order for each person.

Output: Same metrics plus differences across lengths; see whether any knee benefit occurs with higher arm postures or effort.

Trade-off: Best direct answer to the product question, but more trials and force uncertainty.

OPTION 3 | Force-focused single step-down

3 people × 3 conditions × 3 steps = 27 trials

Test: Only if a safe instrumented step/force plate is available. Test no poles, normal, +5 cm; place the landing foot consistently on the plate. Optionally measure pole force.

Output: Foot force + joint geometry → external knee moment → simplified internal quadriceps-force estimate. With pole force, estimate arm moment.

Trade-off: Strongest mechanics, but one step does not represent continuous hiking.

Recommendation: Run Option 1 first. If videos track reliably, choose Option 2 for the final length comparison. Choose Option 3 instead if the lab can supply trustworthy foot-force data.

4 Measurements: what, why, and what they mean

Primary outcome should be an INTERNAL estimate if feasible. Angles and timing support the explanation, but alone do not meet the class requirement.

Metric	How	Why it matters	Be careful
Knee flexion angle (°)	Video pose tracking	Shows whether poles alter the bending strategy during descent.	Measured kinematics; NOT knee loading.
Knee extensor moment (N·m/kg)	Foot force + geometry/kinematics; model	Shows knee-extension torque demand. Lower moment may reduce quadriceps demand.	Modeled net joint moment, NOT joint contact force.
Quadriceps or patellar-tendon force (N or BW)	Knee moment / anatomical moment arm; model	Internal outcome: how hard the knee extensor mechanism must pull.	Approximation sensitive to moment arm, co-contraction, and dynamics.
Foot vertical force (N or BW)	Force plate; calibrated insole if appropriate	Shows whether part of support demand shifts away from the feet.	Force at the foot, not necessarily less knee compression.
Pole axial force (N)	Calibrated instrumented pole (if available)	Shows load transmitted through the hands and poles.	Force direction and pole angle are also needed for arm moments.
Shoulder elevation / elbow angle (°)	Video tracking	Shows whether pole use requires a more demanding arm posture.	Arm posture alone does not quantify shoulder joint force.
Shoulder / elbow moment (N·m)	Arm geometry + measured pole force; model	Secondary internal/external demand estimate: is knee unloading paid for by arms?	A net joint moment is not shoulder compression or injury risk.
Trunk lean, step time, variability	Video + trial log	Explains mechanism and checks whether walking strategy/speed changed.	Interpret as possible confounders, not benefits by themselves.

What we should show in the report

- **Figure 1:** Individual paired points (P01–P03) for peak estimated knee demand, no poles vs poles; no population-level claims.
- **Figure 2:** One example knee-angle waveform aligned to step-down events, with or without poles.
- **Figure 3:** Shoulder angle or estimated arm moment alongside the knee metric to show the trade-off.
- **Figure 4 (if measured):** Foot force and/or pole force over time, showing when the arms carry some of the load.

Simple comparison: percent change = 100 × (pole condition − no-pole condition) / no-pole condition. Show all three participants, plus trial-to-trial spread. No significance claims with n = 3.

5 Test-day protocol (repeat for each option)

01 Decide what forces are available. Ask if the CREATE lab has a force plate, instrumented step, pressure insoles, or calibrated pole sensor. Record what is actually measured vs assumed.

02 Make the stairs safe. Use dry nonslip pole tips, clear landing, and spotter/handrail access. Stop immediately if a tip slips. Confirm course/participant permissions.

03 Set up cameras. Fixed side view at 60 fps if possible, full body + pole tips in frame; optional second phone for OpenCap. Keep positions fixed; use a visible clap for alignment.

04 Record geometry. Body mass, height, leg length, step dimensions, pole lengths, baseline elbow angle, shoes, and preferred lead foot.

05 Practice and collect. Each participant practices 1–2 descents per condition; then record three clean repeats, counterbalancing condition order. Allow rest as needed.

06 Rotate three roles. One participant; one checks video/sensors; one spots and logs. Rotate at each participant change. Cameras must be fixed during trials.

07 Quality check and save. Check knee/arm visibility, pole contact, foot landing, time sync, and raw file. Name each trial P01_C0_T01.mp4/.csv, and log deviations.

6 Modeling rule and decision point

If measured forces exist: use measured force(s) and synchronized joint geometry to estimate a sagittal knee moment; an explicitly simplified internal estimate is $F_{quad} \approx M_{knee} / r_{quad}$, where r_{quad} is an assumed knee-extensor moment arm. Evaluate several plausible moment arms and state the assumptions. If no force data exist: report angles honestly, and use OpenCap's model-derived dynamics only as exploratory, with special caution because contact forces from trekking poles may be missing or wrong.

AnyBody AMMR needs a compatible AnyBody license. HX711-based pole sensing is optional hardware development, not a ready-to-use validated instrument. Do not assume lower knee extensor force proves lower tibiofemoral or patellofemoral compression.

7 Publications and setup guidance

Schwameder et al. (1999), knee forces on a 25° downhill ramp. Measured knee moments and estimated internal forces; relevant precedent, but ramp ≠ stairs.

Bohne & Abendroth-Smith (2007), hiking downhill with poles and packs. Reports lower-limb joint moment/power effects during descent.

Jensen et al. (2011), knee compression during level pole walking. 2.4× pole force did not lower estimated knee compression: a key counterexample.

Uhlrich et al. (2023), OpenCap validation. Supports model use but does not validate pole-assisted stair descent.

Hawke & Jensen (2020), trekking poles review. Summarizes benefits and physiological/upper-body trade-offs.

REI trekking-pole sizing guide (practical, not scientific validation). ~90° starting fit and +5–10 cm adjustment for downhill sections.